

Plan: Cisco Inventory Fulfillment & Optimization — Maple AI

Locked via grill — by Claude + Nabeel Kahwash

Goal

Build an internal decision-support tool for TD SYNnex sales and quoting teams that ingests Cisco CTO quote spreadsheets, maps CTO SKUs to Cisco-vendor-management-approved spare equivalents, queries the live ERP/WMS API for inventory availability, applies configurable percentage-based fulfillment rules, and presents a clear Stock / Hybrid / Drop-ship / Manual Review recommendation to the sales rep before order conversion. The MVP covers U.S. data only. No order is automatically converted — the tool advises, humans decide.

Approach

- 1. Phase 1 — Business Requirements (Jun 8–12):** Document the current quote-to-order workflow, identify all decision points, confirm data sources, and define user output needs (what a sales rep needs to see to make a decision).
- 2. Phase 2 — Design & Logic Mapping (Jun 15–19):** Build mapping table v1 (all entries approved by Cisco vendor management, tracked with approver name + date + Cisco confirmation reference). Define mapping table lifecycle: versioning scheme, effective dates, expiry/revalidation cadence, supersession rules, rollback procedure, and conflict resolution process — all confirmed with Cisco vendor management before any mapping enters the live table. Verify live ERP/WMS API access and confirm canonical ATS semantics with ERP/WMS owners: which fields represent true available-to-sell (excluding serial/lot constraints, open picks, damaged/hold stock, channel reservations, future receipts). Finalize admin-configurable threshold values. **Exit criteria: mapping table v1 complete AND API access verified AND ATS field definitions confirmed in writing AND Strict Match flag ownership and approval workflow defined. Backend dev does not start without all four.**
- 3. Phase 3 — Backend Development (Jun 22–26):** Build ingestion layer (AI parser + human confirmation step for every parsed quote, low-confidence lines flagged and held; explicit fail-closed rejection states for: malformed files, non-U.S. inputs, zero/negative quantities, unknown SKUs, and unsupported BOMs — all rejected before rules execution with a user-visible reason, never silently passed through; duplicate line items are NOT blanket-rejected (repeated SKUs are legitimate across quote sections and BOM components) — exact duplicates are aggregated by quantity, ambiguous duplicates are routed to Manual Review). Build deconstruction & mapping engine (CTO-to-spare lookup against approved table). Build inventory interrogation layer (live ERP/WMS API call with timeout budget, retry/backoff, and circuit breaker; timestamp attached to every result; degraded recommendation state shown to user if API is unavailable rather than silent failure). The tool does NOT perform reserve/lock — it passes a recommendation only. Reserve/lock is owned by the ERP system at human-triggered order conversion, not by this tool. Build rules engine with deterministic scoring: thresholds applied at line-level first (each SKU scored independently), then aggregated to quote-level percentage using formula: $\text{sum}(\min(\text{available_qty}, \text{required_qty})) / \text{sum}(\text{required_qty})$ across all lines in

the build group — this caps each line's contribution at its requirement, preventing surplus on one line from inflating the overall score; warehouse splits counted as available if combined quantity meets line requirement; rounding is floor (never over-count); configurable thresholds: $\geq 100\%$ → Stock; 50–99% → Hybrid; $< 50\%$ → Drop-ship; any Strict Match line missing from inventory → Manual Review regardless of overall percentage. Scoring examples must be documented and validated against sample quotes in Phase 2. Build recommendation output with fulfillment options A/B/C and real-time percentage adjustment UI.

4. Phase 4 — QA, UAT & Pilot (Jun 29–Jul 24): Test matrix covering: parser accuracy (standard + edge-case spreadsheet formats), mapping conflict detection, rule boundary conditions, stale/degraded inventory API responses, concurrent reps competing for the same inventory (double-allocation prevention), recommendation-to-order drift (inventory changes between recommendation and conversion), ERP outage/circuit-breaker behavior, API latency under load, and permission boundaries. Select Sales/Ops pilot, measure recommendation acceptance rate and manual touch reduction. Tagged order cohort tracking begins (followed vs. overridden vs. ignored); overrides must capture a reason code (e.g. "customer preference", "pricing", "lead time", "distrust of recommendation") — minimum viable signal for tool improvement without a full analytics framework.

5. Phase 5 — Deployment (Jul 27–Aug 14): RAMP submission, LLM approval, go-live. Background monitoring agent fires UI flag alerts to sales rep + manager when a drop-ship conversion is detected with available inventory, before ERP order cut-off. Every detected mismatch emits a structured audit event regardless of cut-off status, including: actor (sales rep ID), order ID, recommendation ID and version, threshold config version at time of recommendation, mapping table version at time of recommendation, SKUs with available stock, inventory snapshot at detection time, cut-off window status, detection source, and alert outcome (fired/suppressed). Logs-only path after cut-off still writes this structured event.

6. Phase 6 (Post-MVP) — Canada: Treated as a fully separate workstream. Different SPA file formats, potentially different SKU naming conventions, dedicated mapping table, separate Cisco vendor management approval process. Not a configuration switch on the U.S. system.

Key decisions & tradeoffs

- **Mapping table is a prerequisite, not an assumption.** It does not exist yet as a structured artifact — Phase 2 must build it from scratch with Cisco vendor management sign-off on every entry. This was the single biggest hidden risk in the original plan.
- **AI parser is not trusted alone.** Every AI-parsed quote gets human confirmation before entering the rules engine. Low-confidence lines are held. This adds friction but prevents hallucinated SKUs from generating wrong recommendations.
- **Live ERP/WMS API over cached feed.** Chosen for accuracy. Trade-off: latency and availability risk under high-velocity quoting windows. Mitigation: caching for the recommendation UI display. Reserve/lock at order conversion is performed by the ERP system at human-triggered conversion — this tool does not call reserve/lock at any point.
- **Thresholds are admin-configurable, not hardcoded.** Business owns the percentage cutoffs. Strict Match flag is a hard override — forces Manual Review regardless of overall stock percentage.

- **MCP is MVP-scoped as read-only.** Agents can query recommendations, not trigger conversions. MVP uses long-lived API keys (short-lived tokens are post-MVP) with compensating controls: keys stored in a secrets vault (not in code or env files), IP allowlisting per approved agent, automated rotation cadence (max 90 days), and revocation possible within minutes. All calls logged with agent identity. Autonomous conversion is post-MVP.
- **Canada is not a config switch.** Different data structures, different SPA formats, different approval chain. Treated as a separate build in Phase 6.
- **Tool owns recommendation quality, not delivery outcomes.** Success is measured via tagged order cohorts (followed vs. overridden vs. ignored) compared against the 90% drop-ship baseline. Ops owns delivery and lead time metrics independently.

Risks / open questions

- **Mapping table build time:** Cisco vendor management approval is a dependency outside the team's control. If approvals are slow, Phase 2 slips and Phase 3 is delayed.
- **ERP/WMS API SLA:** No confirmed SLA on the live API under peak quoting load. Needs a load/latency test in Phase 2 before backend is built against it. Degraded state behavior (circuit breaker, user-visible failure message) must be designed before Phase 3 build starts.
- **ATS field semantics:** ERP/WMS "available" quantity may include stock that is not truly available (serial/lot constraints, open picks, damaged/hold, channel reservations). Must confirm canonical ATS definition with ERP/WMS owners in Phase 2 — wrong semantics here produce confident-looking but wrong recommendations.
- **Strict Match flag ownership:** Who sets the Strict Match flag on a mapping must be defined and agreed in Phase 1 (not Phase 2) — it affects rules engine design. The approval workflow is confirmed as part of Phase 2 exit criteria before backend build starts.
- **Intervention window timing:** Background agent alert must fire before ERP cut-off. The exact cut-off time and how the agent knows it need to be confirmed with ERP/ops team.
- **Sales adoption:** 90% drop-ship habit is entrenched. The recommendation UI must show margin impact clearly enough to change behavior — adoption risk is real even with a technically correct tool.

Out of scope

- Automated order conversion (tool advises only, humans convert)
- Customer-facing portal changes
- Complex / non-standard BOM handling
- Canada rollout (Phase 6, separate workstream)
- Full MCP autonomous quoting (read-only in MVP)
- Any LLM model pinning to specific codex/GPT variants on ChatGPT-account auth